ActInf Livestream #036.1 ~ "Modelling ourselves: what the free energy prin

Livestream | *1:47:47*

that was just blues entry music

Hello and welcome everyone to Actim Flab live stream number 036.1. It's January 19th, 2022.

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learning, and practicing applied active inference. You can find us at the links here on this slide.

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improve our work. All backgrounds and perspectives are welcome and we'll be following good video

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participate in any Actim Flab events which go beyond but also support the live streams. And also the

Coda site at the bottom will give you access to the table with metadata on the past and upcoming

live streams. All right, today in Actim stream number 36.1, we're going to be discussing and

learning and seeing who joins for the paper, modeling ourselves, what the free energy principle

reveals about our implicit notions of representation by Sims and Pizzullo 2021. We went over some of the

background in 36.0 and today we're just going to introduce ourselves, say hello, what we're excited

about in the paper, and then we wrote down some things to discuss, but I'm sure both of you have

other random things you wrote down and anything that people write in a live chat we can also look at.

So I'm Daniel. I'm a researcher in California and I think I'm excited to just hear Dean's marginal

comments and just see where it goes because there's just a lot in this paper and so it's just a,

it'll be a thick dot one.

Dean?

Well, I'm Dean. I'm here in Calgary and I really didn't know if I even had time to even look at

this paper and then I, I listened in on Daniel and Blue's conversation in the Zero and that it piqued

my interest and so I read the paper and I got really, really, I was really happy to read this

paper because where, where Daniel and Blue kind of took a path to introducing the material. This is,

this is kind of familiar ground, familiar territory for me, which is very unusual because most of the

stuff that we look at, I'm looking at it for the first time and so I'm really excited and I don't

want to detour anything that you guys want to sort of look at at the point one. Maybe at the point two

point of this process we can look at what the section five of this paper, what can we learn from this

debate because I think that section five of this is super interesting and not just trippy. You guys

use the word trippy, which I thought was fantastic, quite a, quite a good description but I think we can

pull parts of this out and look at them and reassert them, reattach them in our own way and actually come

up with something pretty, pretty amazing. I'll pass it down to Blue.

Hi everyone, I'm Blue. I'm a researcher in New Mexico and this paper, it was great. Like, like Dean was saying,

it was familiar to me but not because of my background, just I thought it really

explained and summarized a lot of like different points in the FEP and active inference and how we

use it to model ourselves and it pulled out a lot of really important questions I think that I'm looking
to, looking forward to discussing with you guys.
So we wrote down some things to discuss but maybe let's start with just a blank slate. Dean, what do you
think would be some approaches that we can take or some starting points because we already have
other cards up our sleeve?
Well, I don't want to be a, I don't want to be a guy holding up a flag and detouring your conversation. One
of the things we might get to near, closer to the end hopefully is the authors speak to the possibility of
you know when you set up your table, Daniel, you, one of the things that they potentially talked about
in the section five of the paper is the idea that the lines are a little bit fuzzy here. Maybe they're
broken and the idea of in some cases our ability to move from the non-representational into the
representational it's kind of a big deal because you can have the debate that says here's the parts
that would align with a non-representational view and here's the lines that would, or parts that would
align with a representational view but then they insert this part that says but maybe we slip into and
out of these representations kind of like we slip into and out of the q density and so maybe at near
closer to the end today we can tap into that a bit.
Let's start with that grid. Yeah. Because this is the structure of the paper
paper and the elucidating contribution and also I think Dave left a YouTube comment or something like
there's other dimensions so this is kind of pointing away towards an interesting approach to
making sense of some philosophical debates kind of laying them out in tables like this.
So how should we approach it? There's the representational and the non-representational
perspectives or stances or modes or archetypes. That is definitely very blurry because of course it
depends on what do you mean by representation and that can have some fundamental ambiguity and then that
representational versus not continuum or dialectic is then separated into a few domains.
I think we could talk about what are the similarities and differences because some of it was a little
nuanced and I wondered if there was other categories or they were organizational structural content related
and functional. So then this is your game board of eight lower dimensional projections that the thought can be in.
Or it's something like one of these eight octants and then a lot of the paper is going through some of the literature
and rhetoric for and against each of these
the FEPs
using the FEP as kind of the interpretant. What is being interpreted to speak for one of those eight cells?
What do either of you want to continue with there?
Well on the action domain so on that sensory motor stuff and
and Yelly Brumberg talks a good argument around the idea that
that for example I could be in a flow state and there is no model that I'm or plan that I'm necessarily
following and that's true. If we're talking about extension and embedded as two of those four E's

you don't necessarily have to move to that place that requires something to work off of. You could just be flowing and working in something right that that that rate of change is is you extended. So there are there are times when

you literally don't have to be working off of a plan. You could also be moving

from a blueprint to some kind of outcome that you want to achieve and then that it's pretty obvious then

that the representational piece has to exist. So maybe the argument isn't that both aren't

at work it's how we transition from a density out of a density and then back into a density.

Something to something to at least have a conversation about.

Because I think Blue was touching on this last time too. She said

I just want somebody to be able to show me how how these lines are in this this sort of structured

relationship exist. Because and I when I when I heard her say that I was kind of like

no I think that's a that's a fair question because those those lines seem to be blurred.

So something that came up yesterday is the idea of free will. And I just wonder how that might play into

these different grid boxes like like is there free will in all of these or in none of them or

this is that even related.

Okay.

Let's write it down.

One interesting thing about this

is it has a little bit of that visual illusion with the black grid and the white dots at the junctions.

So it's kind of a perceptive artifact. It's just sort of messing with our

visual representation and visual representations are are you know when you see it you believe it that kind of

focus system in a lot of the active literature and vision.

So let's save free will and the the blurriness for later and then come to some of the things we wrote down for the previous list.

Okay.

So in what sense can or are

computational models or any other kind of models be fully observable?

Okay.

Can I can I give maybe a possible

real life example that maybe speak starts to speak to that?

Sure.

Okay.

So in the paper he

the author say however these differences matter if one considers structural aspects and the degree of

resemblance between hidden variables and environmental dynamics.

dynamics.

So I was thinking about your question and I so I wrote this down.

If dynamics then the material i.e. the content the structure and the organization is going to have a different

change rate
than the function. So the first the first three columns are going to behave
differently than the fourth.
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happens. You quite literally start and end with a dance between the stabilities and the dynamics
leveraging off of each other. So back and forth. So we need to keep open the possibility that the
stable or the representational and the dynamic, which as Yelly points out, can be non-representational
as well, are both in play. And then I wrote here, because I've used the expression when in doubt,
zoom in, zoom out, when Markov blanket, zoom in, zoom out. So in terms of the computation,
there's actually two computations, one around what we think should remain relatively consistent,
and one which we then apply that consistency to something that we actually want to see a rate
change expedited. So there's actually, I think there's two calculations here to the person
that's trying to figure out how they're going to see what's on the other side of the blanket.
Very interesting. Lorenz and Laplace.
So I like this idea of zooming in and out with a Markov blanket. And I think it speaks to
the way that agents are nested or anything is nested. I don't want to necessarily ascribe agency
or free will to different levels of organization. But I think when Markov blanket, zoom in,
different Markov blanket, zoom out, different Markov blanket. So for me, it speaks to multi-scale
systems and how applying Markov blankets is scale friendly or imputed onto the system in question.
One extension or maybe example of that, we've probably talked before, like the table,
it's in the process ontology, it's material that's tabling, but then that material is also
process based. So sort of this multi-scale process ontology and the identification of a given system
of interest at stationarity, like an object that can be tracked with, but there's objects that are
too fast to be tracked. And there's objects that are too slow to be tracked below the just noticeable
difference. And so within a given measurement or active inference regime,
a given entity can have a attention on a given system of interest. It can't have attention on
things that are slower or faster, like out of its frequency band, but that's discarding 99% of the
system to not consider outside of the frequency band. It's like the observables are just one,
they're very bounded. And so that the slower things are the larger things and the faster things are the
smaller things. So that's kind of a nice... And let me just piggyback on that. And if we're talking
about specificity or scale friendly versus cosmic and scale free, that's also part of that zoom in,
zoom out exercise, because there are quite literally elements of this that are still scale free. That if,
again, as they said, modeling ourselves, we can start from that position of scale free,
and then find the specifics, or we can find the specifics, and ask ourselves, how far out can we

extend our confidence that the stability will hold. So that's, again, piggyback on what you just said,
Daniel. There's all kinds of ways of interpreting zoom in and zoom out. And I think that's, maybe that's
one of the insights that we get when we don't get ourselves hung up on being a really radical, an activist,
even though I'm of the camp that says, we start out in the non representational and activist leaning side,
and then we, the plans and the representations and the effects of that are representational.
Daniel. All right, I'll think of an example that a wet lab example. So I'm thinking of doing
PCR or a wet lab technique where you're not observing perhaps the outcomes of what's happening
for multiple steps. And even then it might be a color change or a number on a screen. And so then
you get limited observables. Those are sometimes your degrees of manipulation, like in a thermocycler,
that's changing the temperature. The only thing you're measuring during the reaction is the
temperature. Sometimes you might also look at like the optical property, but usually just
changing the temperature. And then it's very representation like to have a mental or a cognitive
representation, which can include extended phenotype like the notebook and the computer, but
that's pretty representational to have counterfactuals about what molecules would do that have never been
visually observed. And so science becomes representational. And I think, how far does that
reach back? And it kind of blurs off into increasingly non representational forms. Not saying that science is
the most representational thing, but what are the most representational things?
Dr. But you just move back and forth on that is what I'm seeing with the two feet in, two feet out.
Dr. But you just hit on a really important point because you said science becomes. So what makes up
the becomes? Like what exists in that space prior to it, us all agreeing that it's now science.
Dr. And I think that's what this, I think that's what this modeling as opposed to us as model asks us to consider.
That's a really, that's a really, you touched on the big, the big deal here. Abductively,
so what makes up that, what has to be present for us to be able to say, okay, now you've crossed a threshold,
and now we're talking science.
Dr. Yep. It reminds me of the paper of Brunberg, anticipating brain is not a scientist.
And we can just take the always prescient titles and content of these papers to be targeting
attention point. Like, is the Markov blanket a valid or useful construct? More recently, but back
several years ago, identifying some similarities, or at least a area of interaction with the anticipatory
and cybernetics of systems, which is pretty system agnostic can apply to very abstract representations,
as well as perhaps real systems. And then the physical brain hypotheses and the inactivist and
the ecological perspectives on cognition with this activity called science and the role of a scientist.
And so a few ways to parse science. One is, as we heard from Majid Benny recently,
a model-based science. So that's one like relevant way that we
learned about it and talked about this recently, like science being the application of models,
but then other systems are perhaps doing something different. Well, what is science?
Anyone in the live chat should definitely respond. What is science? And also, I would like to hear what

both of you think and how that relates to anything we're talking about here or ever.
So, so I always think about science as the application of the scientific method to anything,
right? Like, so observations, hypothesis, experiment, etc. And because I think of science like, like this,
science itself is just a model also, right? Like, so, so we use science, it would like, we build a model of
whatever phenomenon it is that we're trying to replicate or observe or test or whatever. And we
do, we experiment with our model, like essentially, so, so the scientific method itself is a model for
doing science, and then the process of doing it is creating a model of the world. So it's like nested
models all the way down in science, in my world.
So, Blue, can I ask you a quick question? What do you think precedes the engagement with what you're
describing? Is, or is, or, because the authors always say, assumptions matter. What, what assumptions do
you think goes into a person saying, okay, well, now I'm going to use scientific method? Are there, are there
things that, that priors that we should sort of pull out and examine that are consistent when a person
gets to a place where they say, okay, well, I'm going to, I'm going to do that?
Yeah, I mean, so, so there's a lot of stuff that's in the scientific, when I say the scientific method,
it's like, you, I mean, it encompasses things like scientific bias, and positive control, and a negative
control. And, and so all of these things are kind of this, in this like umbrella category, when I say
scientific method, and experimental design, like, is not always simple, but, but definitely, like, what
precedes any science happening is simply observation, right? Like, so you have to, like, be observant, and, and
to have observation, you have to have some instrument with which to measure or gauge the phenomena which
you're trying to examine, right? So even if it's just your brain, like, which is probably the most
powerful instrument of all of the instruments, because it's fundamental to any test, a testable
instrument that tests anything in the world. So I think, yeah, like that. So that's a lot of priors, I think.
So, so, is it fair to say, then, that to get to a place of, of adopting a scientific method, one might
first possess before they can actually get into the formalities of that, a field seeking curiosity, and
a way finding bias? And those are two different things, right? One is, I can be a spectator, the other one,
is I've got to get in the boat, I've got to get my skin in the game. Do you think that those two things
are necessary to adopt scientific method or to commit to scientific method?
So, I'd like for you to unpack those a little bit more. So, yeah, can you just describe what you mean
about like the, those two, the way finding and the, the curiosity and... Yeah, the field seeking curiosity.
Well, curiosity, I mean, it's got, it's kind of got four parts, right? There's, there's an epistemic piece,
there's a, there's a sensing piece.
I can't remember what the other two are right off the top of my head. But, but bottom line is, is that
curiosity doesn't just, it's not sort of a monolith. And, and part of that
is that determination because you don't know, but you, you still, you're, you're, you're prepared to
put in some time and effort, uh, to get past not knowing, means that you can take
one of two essential paths you can find out, right? You can go to a source that already knows,

or you can figure out, um, based on sort of collecting and, and curating, um, evidence.
So that, that's the, that's the field seeking piece. And then the...
Wait, wait, wait, wait, wait, what's the difference between finding out and figuring out? Like, to me, those are the same. So like whether you're collecting evidence like from Google or from...
That's epistemic, right? But to, to figure out is to, is to not have a source already, not a single source of truth. That would be the difference. So one, you can find, you can find a truth source and then either believe it or not. And the other is you have to, you have to develop your own truth through, it takes a lot more time because you have to go around and gather a whole bunch of other bits of information that aren't necessarily pre-packaged for you. Does that make sense?
Yeah. And, and also like, what is truth?
Well, right, right. I mean, you ultimately have to be the one that decides that, right? Whether to trust and have confidence in something or not.
Cool. Lots of...
Um, yeah.
Yeah. Really...
Go for it.
Yeah.
No, go ahead, Daniel.
Um, so let's pull one level back from, or two from one and a half fractal dimensions from the curiosity fourfold distinction. There was like the personal factors. So the question was what precedes the type of engagement that might be called scientific?
Right.
And there's some personal factors. They're basically psychodynamic or like situational behavioral archetypes, but they're things about individual, uh, people, humans. What other kind of factors, even if the individual is the nexus of agency, what other factors come into play and like what precedes the kind of engagement that might be called scientific? If the person is our system of interest, what do we, how do we zoom in and out from there?
Well, that, that's my, that's my question. My question isn't, I don't have an answer necessarily, but what I have is, so, you know, before the big bang kind of thinking, we don't know, but is it possible that there actually is something that we can put our finger on that leads someone to have confidence in a scientific method and not just sort of remain in a, kind of in a place of stasis and, and not be curious, right? Because there's lots and lots of people who are kind of quite comfortable just going, going along with whatever is happening around them. But then there's others that appear to have, um, a potential for a deeper commitment. So when, so for example, when, when we were talking prior to Majeeq coming with us, on with us, um, Blue raised the specter that genes don't necessarily have to have time in order to be considered prediction matter expertise, because it has both the cause and the effect included in the, in the structure. So that got me curious, right? Like some, for some people, just making that statement or that, or asking that question, blow right by them, they wouldn't even

give it a second thought. But then for some people, just making that claim arrests them, slows them down, and gets them asking questions on their own. Now, so that's why I'm asking this in the context of, so why would somebody now go from, I don't care, I don't give a rat's backside about science, to, I'm very passionate about science, and I'm very interested in learning this method.
So I think, um, science really is seen by many as a kind of truth finding endeavor.
Uh, um, because whether it's through mining previous knowledge, or generating new information, and maybe knowledge, um, either one of those is definitely like, like a, a, a thinking of the truth.
Um, and in that way, like, aren't we all just looking to like minimize our uncertainty, reduce our uncertainty? So, so I, I look at science, like, like it is an uncertainty reduction task. I mean, that's the, the point of it, perhaps. So I don't know what factors, um, you know, other than the fact that like, maybe we're in a simulation, like, like we could go there. Um, because that, that would like precede the scientific endeavors that we undertake in our truth finding efforts. Um, but, but really, like, the knowledge of science as a way to determine truth, um, I think might be part of it. I mean, and maybe not everyone has taught that. Um, I think about like, you know, in Western culture, we're all like indoctrinated with the scientific method from, I don't know, fourth grade on or something like that. Right.
Dr. And then they give you the PhD at the end when you're fully indoctrinated.
Dr. But, but, but here's the thing, Blue, and this is a, but you bring up a really cool point, and you know, fourth grade indoctrinated. And I think one of the things you and I have kind of gone, we volleyed back and forth is so, yes, science. And so somebody has, has said, this is a, this has got some practical things and some truth finding things, but I've always come back with great science, plan. These, these are material things. My question has always been when, and you bring up the, well, we've decided when it's grade four, but I'm not sure that some, someone who's great, well, let me see that be about nine, 10 years of age is saying it's the same sciences for somebody who's 2930 or 3940. Right. So we can, we can say, plan now. But I'm, what I'm suggesting is, is that before the, the, the plan or the science, there's something before that, some really important things that we're just assuming or willfully ignoring, because now we're onto the plan, right? Because that's what has been reified and, and given the, the, the holy blessing sort of thing.
So, so I think you're, you're hitting something that that's critical and it's why I'm disagreeing with you. So I don't think that the science that you're doing when you're nine is that different from the science that you're doing when you're 39. And the perfect example of that is a Feynman. Like I was reading, I don't remember, maybe it's in, in the meaning of it all, like maybe it's in that book, but you know, he talks about how, like for him, he just plays. Right. And so, so for a lot of his science is like, oh, I get to like make this toy and play with it and try to look at it. And I keep playing with it and I keep playing with it. And so you, this is something I think that's like innate in us, like to just the figuring out part, like the truth finding part. So even as a baby, like from the time you can sit up, like, what does this block, like, what is this shape? What is this slinky? Right. Like it's finding out and figuring out like, what are the affordances of the objects?

Right. Right. And, um, some of them are imposed by the outside. Like don't stand on the chair. Um, cause kids will use chairs as tents and step stools. And you know, like, so it's some of it, like that we impose some structure on, you can't use this toy in that way. But, but really like kids will, they don't know, they'll use anything anyway. Like they'll, they do all kinds of crazy things. Like, like they'll, you know, slide down a mountain on, you know, a trash bag or something. Right. Like, so, so they don't know, right. How to, what, what the limits are. Um, and so they'll explore new limits. And I think that scientists, um, at least the ones that really love to do science, uh, play a lot also. So, so again, the, the, the, the trial and error piece can be like the triangles piece. It can be scale-free and I would never push back on that piece. The epistemic and the specific is what you just described. The other two parts of curiosity are the distributed and the sensing. And that's kind of the sort of the more act active part where you have to kind of go out and do the skin in the game. You have to get in, get in the boat and cross the South Pacific. You can be doing the epistemic and the specific as well, because you can be taking samples, but you can get, you can kind of see that there's another side to this. So my, my question around the, when isn't around the, the, um, the ability to see the scale-free part of it. My question is, is if I'm a 10 year old and there's something before the plan, how sophisticated is that something relative to the something before the plan as a 40 year old? I just don't think that the experiences that a 10 year old has and the resources that they necessarily draw upon are the same. So I do think there's also a specific and a scale friendly part to this, would you, would you push back on that?

I mean, I think that we all kind of just operate on whatever level that, like, I, I do agree like that the knowledge of a 10 year old, isn't the same as a knowledge of a, of a 40 year old, but, um, I, I think, you know,

you know, we just operate at whatever level that we're at. Similarly, like a baby, you know, like they're just doing their own kind of truth finding, but so the process is the same. Like, even though the priors may differ, I think the process is similar.

This reminds me of in the linguistics guest stream with Elliot Murphy, how there's the view of language as the construct, like subject, verb, object, and then there's the language as learned by each learner. And so I'm almost hearing like blue saying it's the same process. Everyone's on the same trip, but of course recognizes that each learner is going to have individuated factors, including like what they specifically know, maybe has some correlations or, um, whatever with age or with other features. And then that's, I feel like maybe partially reflected with a specific of the curiosity. Um, like if, yeah, is the nine year old and the four year old learning the same science? It's either pan-scientism or uni, uh, solipsism, science-ism. Everybody's in their own scientific niche. And so it's just a million different granules. And then the other hand, the other end of that extreme debate would be, it's all one integrated extended cognitive process, which I wrote some notes on. So Dean go for that. And then we'll think about it as more of an extended continuous process. Yeah. And again, I'm not, I'm not trying to, I don't want to sort of shoehorn something in here. And, and, and, and so I would always, I would always hear from people in my history, um, the expression, um, well, I know that's a problem there. I guess you're just going to have to get creative. You're going to have to think outside the box and being the, um, insurgent and the non-compliant person that I am, I would say, I would come back with, and I was reporting to people. I would come back with, so tell me about this box. What makes up the box? What makes up the process? And of course that didn't make me very popular because nobody could get specific about what, what's inside or what makes up that three-dimensional space. And, and I, I think it's the same thing with, we can say, well, there's a process. Well, tell me then why a 10 year old thinks in the same way as a 40 year old does what evidence, what, what, what, what are you hooking that assumption on? And again, I'm not, I'm not trying to be a pill here. I want, I'm really curious what would make a 10 year old think the same way as somebody with an additional 30 years of experience. They want uncertainty reduction. That's a generalization that I can agree with, but what about the specifics? I'll give a specific in a general. I think that for disciplinary education, some of the questions might be very similar. Like why is this leaving group superior to that one in organic chemistry? I think if somebody were very precocious and 10 and learning about the halides as leaving groups, or if they were learning about it in adult education, they would ask that question. And then once they got the answer, they would have moved on in their learning in that disciplinary finding out way. Like here's the periodic table. Here's why the trend is this way. So that's not the whole of education, but in that mode, which many of us have come from and exist within that was the education that's educational instruction. So that's the instructionism component. And then I think it becomes different as you move away from that spotlight. So then a few other comments, like about the specifics of how the different age would be different. There's the, the biology of how the individuals would be different, like in some of their cognitive features, whatever it meant for that person from working memory to the specific long-term memories and other capabilities they have, perspectives they've learned about. They're a different complex system, but they can be a complimentary part of a system like older people providing mentorship for younger people. But I think that's what kind of comes to some of these notes on science as an extended cognitive phenomena, which is partly what has been got at. So science has many different possible goals. And sometimes scientists in an ad hoc way will appeal to different components. Like science is correct because there's technology, like it's demonstrable or it's useful. Not always providing a null hypothesis for what we would have if we had done something different, but there's all kinds of different ways to look at it, but it deals with individual cognitive features and learning and action in their niche. But it also refers to broader phenomena. Like maybe a differentiating question is, can one person on an island by themselves do science? And if the answer is like, yes, they could explore and hypothesis test, then that's one perspective. Another perspective would be you're only doing science when you're using this interpretation of the scientific method. And you're using science as a body of knowledge. Like you're using the literature with the DOIs. And some people go as far to say that you need to have the career in a certain specific institution or a specific degree or a specific training or approach.

So that is clearly not applicable to the one person on the island. And not that one of them is better,

but that whole apparatus that's part of the extended niche. And then also there's, at least in our case,
the emphasis on the digital or at least informational stigmergy. Like there's a recognition in science in a
different way than like, there's many books you could read on tying knots, probably a lot of YouTube
channels. Is there a DOI for information on knots? Probably not. But that in science is a big emphasis,
reading and writing the literature, at least in the way that we know it today. But again, that could also
perhaps not be as much of a core component. Anyways, just like a few points that this,
maybe now we'll bring it back to this paper, but science is kind of the representational paradigm.
And I think Dean's question is getting at what is before, what's the before and the after,
or the subconscious or the submerged of science.

Dean's question is, what is, where is, how is when we set up our hypothesis. And I think what this paper,
if we're bringing it back to the paper, I think what section five of the paper speaks to, especially
when we're talking about the functional dimension is the if, the counterfactual, and how they talk in
there about how, I'm not sure if it was Clark or how, how here, one of them talks about the fact that,
so a bacteria doesn't typically ask the question, what if, when it's sampling its environment. So that's,
that's pre science.

Yeah, I think that relates a lot to Martin Boots is guest stream, I think 12 on the event based cognition,
and how if there is an event based structure to cognitive events, then for any given parameter
in that model, it can be otherwise. You could say like the ball is rolling off the table,
the ball is not rolling off the table, the cup is rolling off the table, like you can start to
just do single mutations, especially the important counterfactuals for linguistic structures.

And then that might be closer to what we've seen discussed in other SIMS papers as the adaptive
active inference agent, going back to earlier papers citing that, but just that was our previous
discussion with SIMS. This paper has SIMS, like the pendulum does not have a counterfactual about whether
it could go on a different axis. It's mechanically constrained, whereas cognitive structures can
engage in counterfactual play in a way that the pinball machine can't, even if the pinball machine has
chaotic or other types of seemingly agentic behavior.

And in that space, I can be at the top of the ski hill prior to a gradient descent,
and I can be adaptive, meaning I can close my eyes and do a visualization, and I can also be variational,
meaning that I can try to recreate through my proprioceptive sensors, a sensation of floating,
pre-pushing off and engaging with that process, right? So again, the when of this, I think, is something that really matters.

I think we gloss over it, or we don't even pay any attention to it typically, because we want to get to,
in the paper, it's figure two, we want to get to figure two. But I think before there was a figure two,
before there was a plan or a representation, we might want to ask when plan. If I'm engaging in
counterfactual inference prior, then doesn't everything in the diagram depend on prospecting
up the nodes in that representation? All right, do I not have to provision up a format that will be
potentially something that others can comprehend as now organized? And don't I have to propose
alternative paths as links? Like all of those things are pre-planned. All of those things have to be

present pre-science. So we can say, well, I'm not paying attention to that now because, you know, I'm busy putting processing something that I can now transfer to other readers and viewers of this diagram. But I think to ignore those things is to ignore the difference between a 10 year old doing a science experiment and a 40 year old doing a science experiment.

Let me give one thought on that. It actually speaks to this question of whether science is the same for everybody or different for everybody. Is this the board exactly as laid out that everybody has to play by? That's like the game of life, the board game, not the cellular automata simulator, but the board game, there's one path or there's limited points of agency. And then everybody on the board on that layer one is moving through life on the same or on a constrained set of trajectories. So it's like, are these the immutable railroads of decision-making and thought or might other people have different connections or different decisions or totally different railroad stops or different dimensionality or just represented in a different format? Is it like, well, here are the battle lines that are always going to be the case for philosophy. And there's going to be some people whose situation and neurodiversity leads them to be on this side or this quadrant or emphasize this component, the regime of attention, or is even that layer shifting because there isn't any stability to be found in counterfactuals.

There you go.

Back to the questions.

It was a great early point about the rates of change being different because also that does, at least talking about rates of change, kind of evoke numbers, gets the scientists excited because could there be like a number? Could it be a sentiment score? Or could it be like this sentence is totally consistent with both or it's inconsistent with this? Could we annotate the literature and measure or do surveys or actually measure those rates of change in individuals on their learning trajectory or in fields? Like how could it go from just being a really insightful philosophy paper with a lot of scholarship into being something where we can use some numbers or something?

So I see the first three columns organization structure and content as the more stable.

I think you had another diagram where you were showing that going out over longer periods of time, right?

So which one do you think is the most rapidly changing? Functional.

Okay. So let's kind of recap them. So organizational seems to be related to the encapsulation of the organization, like the architectural features of the world, such that the representation is separated from the, what is represented. So like you could have representation of the sun, but then that's very far physically from the sun. The structural aspect is related to representational vehicles to that are structurally similar to the state of affairs in the world they stand for. So that is the abstract structure of the representation, having some congruence or isomorphism or like similarly, some similarity to what is being represented. This one is super dependent on the definition of representation.

Cause someone go, well, you can't represent a car in your head because your head's not a car content related, having internal models that either encode environmental contingencies or sensory motor contingencies, specification or description of how the world is taken to be turned analyzed in terms of correctness or truth. So that seems to be more consistent with the car content. You know, it's like the

YouTube content.

The car content can be represented if it's stands for something, the content of that message was an invitation to the birthday party.

So that is related to the information or like the memetics, especially in a actionable way.

And then the functional role, this is where we're going to highlight and hopefully Dina, I want to hear a little more about why this one is different.

It supports vicarious use before or in the absence of external events, of internal variables of model.

That was discussed in the paper a lot in the context of sensory motor detachment.

Yep.

And there have been some blurry lines found there for sure, like motor replay and pre-play.

So we know that it is on the spectrum of existing that neural dynamics, but cognitive dynamics involved during

are involved echoes or anticipations of them are involved in the before and in the after, as well as counterfactuals before and after.

So what is happening with functional representations and then maybe how does it relate to active and FEP?

Yeah, so I don't, I don't know, I know enough about dreams and subconsciousness just to get me into trouble.

But the first three, when we reorganize through our dreaming and whatever those representations are,

non representations depending on your, your wish, those are all backwards looking, typically.

And then the functional, that's what segregates it out is it's forward and backwards.

So that's to me, a huge differentiator right there.

In terms of how much, how much, or how long we expect something to stabilize versus how much we want

to act on something and, and generate that change or outcome that we're seeking.

That's why, that's why I get back to that search field being different than a framework.

Okay. It's a lot to think about, but it's interesting.

The organizational, maybe there's some dissenting view from this, but this is related to how the system is arranged. So this one is, can be thought to be slowly changing because we're talking about things, which means we're talking to something that has persistence. And so the organizational aspects of systems tend to be the slowest changing, perhaps like the organization of the servers changes on a very slow timescale relative to the processor.

Then the structural aspect. Also, it's very important that it has persistent, like storing a variable in memory. Like you want things to stand for each other for a long time. You want the DOI to be a representation of that paper for a long time. Those are also related to like, it's sort of the functional side of organization, like form and function a little bit, but I know for functional is used later.

So not to confuse the term function, but this is related to some of the performative aspects of organization and infrastructure and architecture. So that's also like, that's, you don't want the screwdriver to be a different thing in your hands. That's too slow, but maybe it'd be cool if you could modify or tweak the tool or like, you know, unscrew it and bolt something in a little slower to make it different, but usually you don't want to change too fast then. But what about the content related?

This seems to be also changing somewhat faster because environmental or sensory motor contingencies, that seems like that could change on the fly. And then I agree that the functional role as defined here, like vicarious detachment, including counterfactual and self modeling counterfactuals. This is like some of the most rapidly mutating, but what do you think about content also changing fast?
Well, I think whenever I make a mistake and I accidentally hit the delete button on my notes, how upset I get because the stability of the content is lost. So I actually see it is still relatively stable as opposed to something that's changing rapidly.
So these four columns here, there is a different temporal aspect, which relates to what you asked earlier about when representation. So instead of the possessive, do organisms have representations or the identitarian are organisms representations or the definitional, what are representations? This is the temporal or the dynamical when are representations? Well, that's not complete.
You have to say what it is. You can't just say when it is. Well, couldn't you just flip the words?
Wouldn't it be just the other person's perspective? You can't talk about is without when. So why is that one?
Okay. Why are there papers about what is a representation? What is it like to be a bat rather than when is it like? I don't think it has to be or I think it's both. The other thing that I would I think we need to maybe think about that is the first three tend to be on a continuum that's orthogonal to the fourth.
I can organize, I can structure and I can content build. Functional now is orthogonal to that. I will because it's going, it's used before. So again, back to that temporal aspect. It's both. I wouldn't drop any of the four columns.
I'm just saying that I think that the fourth column is maybe on a y-axis relative to the first three that are on the x plane. Okay. A few notes here. So before or in the absence of. Right. Now, isn't it the case that usually anticipation entails something being both before and in the absence of? I wonder what it will be like when my friend comes over that's in the absence of the events and it's a counterfactual before, but it may happen. It's a prediction. It's an expectation and maybe even an active inference. It's a preference.
So I agree. Like even the way that the authors have written it, these are related to architectural and performative features of like basically functional or structural definitions of representations, what they are, what they do. And then this is even uses temporal words as well as counterfactual words though. And those are probably like used elsewhere in the paper, but that's a very interesting distinction. And so it's like, there's our map, but you know, how easy is it to just lay out the map and go, all right, here's Northern California. Here's a map instead of the temporal aspect.
And I think that also shines a light actually on another feature of what people would call science, which is like science, the metric system. And then like, okay, you've gone too far with the metric time, but then pretty much on a lot of the other fronts, except in the United States, um, the metric system has, um, hold as how people are sense-making in terms of like distance and time and other units.
And that's chronos like, like, like, um, chronology and time in the decimal time.
And then kairos is like the timeliness and there's other ways to think about time.
So we kind of open it up and it just, um, makes me wonder, okay, what else is handed with the map?
Is there a time object? How do we share time along with the map? Is that attention? Is, um,

focus times, time, times, all these other things. So how do we convey the temporal aspect and instead of just this gene regulatory network, or this is that the map here or the map and the math and the territory, how do we bring the temporal along?
So I wonder like about the temporal also, and especially like now that we're kind of delving into, um, like quantum aspects because you know, like the, the simultaneous existence of more than one fate, right, is possible. And so is the, the, is it possible to be like building and like executing your model at the same time? Like, can, is it input then output? Is it always sequential or, or is there some kind of like simultaneous happening, right? Like that's, that's going on there? I don't know.
Oh, blue. That's perfect for the next question I wanted to ask. So is one of the assumptions here that there's a, a continuum. Like if then between I can, and I will, because this was a debate that I got into with people forever, when I was in education, that there's some sort of an assumption that because I can, therefore I will. Well, I wouldn't jump off a cliff, even with a parachute on.
There's no will for me to do that. Cause that's just not my, it's not my jam. But, but it, orthogonally speaking, if, if, if organization is on the, on the X and structural is on the X and content is on the X, but functional is on the Y. Now I can separate the two things out. And, and, and as you said, if I go back to the quantum level, now I'm, I'm pre decision branch in that diagram. Do you notice that the diagram is basically unidirectional? It's always heading to a particular outcome. It's always, I can make this decision branch moment. Therefore I will arrive at representational or non-representational. And I'm with you. I'm not so sure that previous to this, that it isn't bidirectional. It isn't functional. It isn't orthogonal. I think it could be all three of those things.
Lou, would you like to say anything or can I? All right. Two points I would be remiss not to make.
First, the tetrahedron and fuller synergetics gives us a mnemonic to have four spatial dimensions.
So this four column scheme, all of them can be four intersecting spatial dimensions.
They're coordinated at 60 degrees, not 90 degrees. And then the second point is like this one directional maze reminded me a lot of the way that people study decision-making and path finding or figuring, whichever one may be in ants, which is there's some laboratory maze, like a Y maze or a T maze, or maybe some branch on a tree in nature. And then for the purposes of the science, so-called the paper, it's going to be like, we counted them as going down a certain branch, making a decision at the node when they went five centimeters past the branch point and didn't return. So it's like, this is the, the line that you trip when it goes from zero to one on that number. And that's what gets fed into the scientific process and the method, the observations of the literature and the statistics and all of that.
But then if you're watching the ant, or if you look at the video, maybe they dwell at the node or they turn around or they walk down one and then they come back or like, even they go far down one and they come back. So it is very interesting and it's a subtle graphical point, but one that would be almost seen as like, what do you mean? How could these be undirected edges? Well, what if we say, hey, let's start at representational FEP. It's just where we're going to start today. And then let's pull back to the vicarious use of variables. Now, where can we go from there to sensory motor detachment or we can take no. And all of a sudden we're a non-representational FEP because they're

only one step away because they're both outcomes of the decision about the vicarious use of variables. So I hope those are some interesting points. And I think that relates a lot to the discussions that we've been having in EDU about cyclic curriculum. And you touched on it. I'm going to just bring it, I'm going to reduce it, not over reduce it. I'm going to reduce it to your point, Daniel. There can be a yes, there can be a no, and there can be a maybe. And the maybe part is what's being left out of this.
Cool. I'll read a comment from Serval in the chat, who I was also thinking of when we talked about like science as a human phenomena and all of that. I think Serval has a lot to say on that. Serval wrote, considering many world, Hey Dean, I'm going to read Serval's comment. Considering many world type theories, is it possible that can entails will? It would be a distributional property though, not a trajectory property.
Okay. I have to think of that in terms of what I, how I set up entailment.
Maybe. I'm not trying to duck. I think there will be times when my confidence through my ability to say that I've done something before leads me to a lower threshold around will. But then sometimes it doesn't. So I'm not, I'm not trying to be evasive, but I can think of times when it can entail. And I can also think of times when it very much does not.
Daniel Schroeder So will,
it's a short English word. It's could refer to something that will happen. Like the ball will go to the bottom of the bowl, unless anything else stops it from doing that. Or it can refer to sort of that motivational psychological energy, which is related to agency. And so it's real, it's kind of an interesting, uh, congealed word because agency and will are very related. And, uh, I'm just trying to think if they're positively related, like to feel that you have the choice is to have the agency and that can be seen as willpower. Whereas willpower doesn't come into play in situations where there's no agency. If you just need to sit in the seat on the airplane, it doesn't require any willpower agency for that action to occur. So I'm just thinking about survival's comment and um, the distributional property like counterfactuals. Do we engage with them as a possible, uh, cosmological reality, like in the many world type theories or just, uh, cognitive possibilities so they can be engaged with without taking on like a cosmological meaning. And then what does that mean about our n equals one trajectory, the specifics of our kind of entity in motion and then the distribution of perhaps different types of counterfactuals or different kinds of possible stochastic rollouts of the system given really similar starting conditions?
And that's where the will can entail the can. I can start out by asking what if, but I still have to select the can, the which if, so that's what again I, at some point we can have, we can see a relationship of entailment, but I think as, as a, as a quantum question, they're orthogonal. Can and will are not one leads necessarily to the other.
So just to kind of like play on words and loop back around to what I was talking about earlier, um, like can is an affordance, right? Like, so if you have an affordance, like, like that doesn't mean that, that you'll use it right. And will like means I, I will choose or, or implies that you have some agency, right? I, I choose to, um, you know, not start a fight with a flight attendant while I'm

sitting in my seat on the airplane or, or whatever, because you can do all kinds of things. Any, any,
you get many affordances. Um, and I just wonder how this ties into these representational and
non-representational aspects of, of the FEP. Like really when I was talking about free will earlier,
if internal states are a direct representation of external states, it seems to me like I kind of feel
like there's less options there. Um, whereas internal and external states have coupled dynamics
that kind of leaves some like mystical fluff outside, like that I can potentially leverage with my own
choice or, or something like that. I mean, I, I don't know what to call it mystical fluff, but, but, um,
it leaves something unsaid, right? Like there's a coupling, it's like a hidden, there's a hidden
states there that like are only, that that's where I can impose my will. Well, I'm going to put that fluff
on a pedestal. I'm going to honor the fluff because I do think that there are actual times when the,
the, the ability not to be captured by our observations is actually an advantage. So mystical,
meh, maybe uncertain, yes. And, and is, and there, are there, are there advantages to that at times?
Yes. So we can call it fluff, but I think sometimes, um, we don't pay it enough respect.
I think that relates to the cloud of unknowing, which is perhaps a text if somebody wants to go
into a more esoteric Avenue, but I'd like to return to the figure and ask what either of you thought
about figure one, how does that play into this conversation we've been having? Like, what does
this entity and action partitioning have to do with science? Is this partitioning unique to active
inference or other partitioning is possible? Does it, could it be another way?
I think this is a pretty stable representation and it, and I think it, and I think it does honor to the,
the, um, scale free aspects of what the free energy principle is pointing to in terms of a,
in terms of a, a statistical pathway.
Yeah. I mean, how many debates and uncertainties could be prevented by saying this is a representation
of an entity and having people know what was meant by representation.
Right.
Because it's like, I made this representation of an entity. I think it does this useful thing.
It's like, those points cannot be false. They're assertions by a person.
And what would be disagreed with that? It's not a representation. Someone could say there's a more
useful representation or more good, true, beautiful, simple, but then let's see it. And then we can use
science to compare representations because that would be okay. Right. And then you said it does honor to the
scale free. So the only thing that kind of hints otherwise is the person and the world kind of gives a scale,
uh, indicator for this model, but without that iconography, it would, uh, be it. And also adding
in perhaps a legend or description of what the edges mean, because it does say schematic of reciprocal
exchanges and it uses statistical vocabulary, but it also uses actual like ontological vocabulary
in the terms of actuators. So are these the physical things that are doing the actuating
or are they the statistical or informational exchanges, or is this the representation,
the form of the scientist model? So we're only ever talking about scientists using these models.
So we're only ever talking about that kind of nexus where we're referring to both loosely. And so it

shouldn't be interpreted as either of them specifically. It's just dialogue. There's a lot
of ways to interpret what the edges are and, uh, they might have similar different labelings and different schemes, but it is statistical. It is quantifiable.
Um, it's amenable. It's a qualitative topological partitioning that's amenable
and beyond to certain quantitative analyses. Like if you can draw it like that, you might be able to do a Bayesian graph and then there's software toolkits and there's implementable algorithms. And then there might be some quite interesting guarantees or tendencies with certain architectures, for example, to perform predictive processing or to have learning anticipation of memory or to engage in control or anticipatory control. But that's definitely some of the most core iconography that we've seen.
I mean, I mean, 50% of papers or what we'll find out blue have the action loop
because
what a little side hustle of you blue got going here.
That's funny.
Okay.
Sure.
We're trying, we're trying to, um, read more papers together
and learn about them. Okay.
But it's interesting how often this has come up and sometimes authors draw on the same image.
Other times they do different. And sometimes the arrows are just the clock, like just a
uni-directional flow. Other times there are some bi-directional arrows and, um,
the, there are several guest streams on the more technical aspects there, but, um, let's return
to the general questions. Anyways, anything else to say on figure one though? Like just on this,
like, what is this and how does it matter?
I think this represents better the, the eye, eye fixating on the, on the pot fly than it does on
the physics model. And so I think it's until somebody, as you said, until somebody comes along
with a better representation, one that, that seems to give us the maximum amount of information that we need versus the maximum that we can handle. I think this is a, this is a good rep.
Great and very satisfying conclusion.
It's yeah. Go ahead.
Sorry. Uh, I, I really think that there, um, what's not shown in this representation and what I feel like is always left out is the fact that the, the, there are some states are hidden, right? And so like, where is the hiding here? What is hidden from what, um, in this? And so like the sensory states, we don't even know if like we perceive, I mean, we know that we all perceive temperature differently. Like some people I'm cold in this room, but many people would be comfortable. And some people like my favorite color is purple. My daughter's favorite color is yellow. So like, we don't perceive things in the same way. And so there's, there's always something hidden from everyone else. And I think that this, this schematic kind of leaves that out. I've seen the, the hidden states, um, represented well in other, like action perception loops. But I think that it's important to, um, recognize like what's obvious

and what's not.
If I could give a one thought there then Dean. So I think there's, there's two levels of like,
what are the hidden states from the perspective of the internal states, the system of interest,
the external states are hidden states implicitly or explicitly modeled, but they're the ones that are
never directly observable. They're only observed through their proxy sensory states and enacted,
or the transition frequencies are modified through policy selection through action. So within the
model, within the tetrahedra, the hidden states are relative to the internal states on the other side of
the Markov blanket. And then there's like the implicit tetrahedra, which is like us modeling this
fourfold partitioning. And then everything that's not included as a variable in that statistical model
is like implicitly a hidden state. But then what is that? But then we'd have a different model if it
wasn't a hidden state.

Dean Mitchell, Jr.: And the other thing I think is that the vertical line between sensory states and action
states, if you tip that 90 degrees, it would look like a blank, even though it has two points on the end of
it. And I think that's that blank is by definition, hidden state, it's to be filled in. So I think, I think, again,
you can, you can fill in the blank, or you can leave it blank and say, because it's blank, that is by definition hidden.

David Pricey

Dean Mitchell, Jr.: And that's the sigma, or the mapping function from number 26 and number 32,
a mapping function of internal states to external states. Now, look at Axel Constant and Jelle Brunenberg's
work to think about how the niche is also doing like anticipatory modeling of the individual. So the other
way, but let's just focus on the entity and its mapping to external states. It's predicting external states.

David Pricey, Jr.: Or at least acting as if or can be modeled as if. So there is an edge there that we've seen in modern work.

David Pricey, Jr.: It's not drawn here, which actually might suggest either the inheritance of a legacy graphical pattern or some

preferences or conclusions about what these edges do represent, again, even though they're unlabeled. So that's one edge that we've been hearing more about. And then I think another edge that Dini just highlighted is this one directly between sensory and action states.

David Pricey, Jr.: And again, the clock flow doesn't have this. The one that's just internal action, external sense, that doesn't have the sense action bridge.

David Pricey, Jr.: Right.

David Pricey, Jr.: And it depends a lot on what these things mean, but it'd be cool to understand what is meant by the backwards connections as well as by the vertical connections. And why are there not backwards connections between action and internal and sensory and external? It's like, it's really a four by four matrix with some connections. And that's how we've seen it.

David Pricey, Jr.: And that's how we've seen it represented as the sparse connectivity matrix in number 32. And thinking about the way that we model complex systems and generate approximations that sometimes can also retain some other features like the highly operative exponent. So some thoughts there, Blue?

Blue Pricey, Jr.: So we've definitely seen internal states and sensory states like that reciprocality happen.

Blue Pricey, Jr.: But I do. I always wonder why, like, especially since we did the mental action paper, like how action states don't influence internal states. And so for me, like, when you take action, like when you practice playing the piano,

Blue Pricey, Jr.: That affects your internal states totally like not just through external states or even mental action, like where you're doing nothing with external states, it's entirely internal.

Blue Pricey, Jr.: And so I always wonder why that action and internal is there, but I less so wonder why there's not a reciprocal loop between external states and sensory states like what, like I never want the sensory states to go back to the external states.

Blue Pricey, Jr.: For me, that doesn't make sense. And maybe you guys have an example where it does make sense, but I always want the action and internal states to loop together to have it bi directional.

Blue Pricey, Jr.: Here's one possibility on the sense back to external, but it depends, of course, on a specific interpretation of those, a photon.

Blue Pricey, Jr.: Let's just assume it's out there. It hits the retina. If we talk about the retina as a sense state using it very loosely, not the statistical variable correlated to the sense state, but the actual retina, it's the absorption of the photon that changes the niche to be sensed.

Blue Pricey, Jr.: Like if you're going to be sensing molecules from the ants cuticle, you have to take some.

Blue Pricey, Jr.: You could sense from the ground, the ones that have already been deposited, but if you want some from the thing, you have to take a sample.

Blue Pricey, Jr.: And so that may, whether that would be better understood as being an action in the niche is another question, but like you do have to change the external state to be half skin in the game.

Blue Pricey, Jr.: No. And I just thought of it as you're saying that like the observer phenomenon, right? Like, so there's that too, right? Like just the, just the act of sensing it changes it.

Blue Pricey, Jr.: Or is the observer phenomena. Does that come from sense? Does that come from internal states? Does it come from action states? Are the observers just acting in a way that might fit in within an action state? But those are really interesting questions. Dean?

Dean Cade, Jr.: No, I just think you're right. I think both of you are here.

Dean Cade, Jr.: We're getting, we're getting closer. I don't think we'll ever fly directly into the sun, but we're getting warmer.

Blue Pricey, Jr.: Yeah, this will have another few minutes to talk. This has been a really great discussion and I look forward to like re-listening to it and discussing more. Then of course, feel so appreciative that we'll have a dot two as well.

Dean Cade, Jr.: So we can return to some of our early ones. Free will, maybe save it for the dot two or if we want to talk about it now.

Dean Cade, Jr.: I think we've blurred and fuzzed some of these lines a little bit. So we did kind of get there. Let's see what else we had down. And of course, if anyone in the last 30 minutes has any questions they want to write.

Dean Cade, Jr.: Right? So Steven wrote, I like your method to work backwards in the flow diagram. Start at representative or non representative assumptions, or perhaps conclusions like starting at those endpoints as assumptions, then looking for prediction errors in our action perception.

Dean Cade, Jr.: This maintains uncertainty in ways that does not happen when following the causal effective correlations.

Dean Cade, Jr.: That's sort of like 20 questions is framed as a game of, oh, I'm sorry, I didn't know that mentioning common children's games was funny.

Dean Cade, Jr.: That's framed as a game of going from the most uncertain towards more certainty. Like if you were asking uninformative questions.

Dean Cade, Jr.: What are you doing? If you're not, if you're just getting like, and that's information theory kind of exemplified, you're reducing the set of what you can be guessing with the preference to like win within 20 questions. And then this is, as Steven is pointing out, kind of the other direction.

Dean Cade, Jr.: But then what does that look like? And that is almost inflating uncertainty, at least in the information theory sense.

Dean Cade, Jr.: So here's Steven's question, and then either of you, if you have a thought on this.

Dean Cade, Jr.: Do we model the temporal lags in dynamical change between sensory state in relation to changes in action state?

Dean Cade, Jr.: So maybe a related question would be how do timescales and temporal lags play into our modeling of action perception and cognition?

Dean Cade, Jr.: This is why I kept saying to young people when in doubt, zoom in, zoom out, and it wasn't just zoom in, zoom out in a sort of a visual sense.

Dean Cade, Jr.: You have to zoom in and zoom out on a temporal sets as well. You have to think about the unit of analysis, both as an object, but also the length of time something can remain stable or not.

Dean Cade, Jr.: I don't think we, I don't think we're naturally inclined to include the timeframe as much as we're inclined to include the, did it change? Did it float? Did it burn? Whatever. Whatever it is that we're waiting to see what happens next. We don't tend to give as much attention over to how long did it take?

Dean Cade, Jr.: I'll give a complimentary thought, which is more related to the kinds of models that we've specified.

Dean Cade, Jr.: So the deep temporal models or sophisticated active inference in, for example, a three time step model at time step one, it's doing inference on one, two, and three. That's planning as inference and anticipation. At time step three, it can still be calculating the model in light of incoming evidence about time step one, two, and three, which is related to memory and learning and all these other sorts of so-called

Dean Cade, Jr.: Backwards looking types of inference. So there's like anticipation, there's now casting, and then there's reconstruction of your recent and long-term past and how that's related to identity. So I think that the question, do we, or how do we model the temporal or just the temporality of sense and action states? The answer from model based science is what's the time horizon parameter on the model that we ran?

Dean Cade, Jr.: Oh, it was 15 days. Okay. Well then we modeled the temporal lag up to 15 days. And that is what we modeled. And that is sort of where the scale free becomes the scale friendly becomes the scale specific. And then that's the, the most that it contracts down to is that one specific model as deployed by those researchers.

Dean Cade, Jr.: Here's another question from Steven. Here's another question from Steven.

Dean Cade, Jr.: When do we sense and when do we act? When do our sense states update? When does our generative model update? And when do our action states update?

Dean Cade, Jr.: I'll start with the model based science one, which is, again, it would just be the specifics of how the pseudo code and the actual computer code were written, which line of code updates first? That's the narrow answer, but what about a more broader answer?

Does it depend on how open we are? How confident we are in staying in an uncertain, non-specific space? And can we necessarily know going in how long that will be? I don't know. I don't know how long. And what about precognition? If we expand to that possibility, I knew that was going to happen. I mean, that happens to all of us. We all experience that, like, oh, I knew that was going to happen. If I step on this ice cube, this ice pile, it's going to crack through to water or whatever. I knew that was going to happen. So we're able to perceive things before they happen. And so I don't know, like, is there not necessarily the temporal separation? This makes me think back to Shanna Dobson and the flattening of the time diamond. So really, is it necessary that there's sequence here or can it just all be happening simultaneously? That kind of relates to what I was asking earlier too. Like, is there some kind of simultaneous update sensing action possibility? And I think that there is. I mean, I don't think that, not in a computational model because, you know, code just runs like it is, but I think in our model, it's possible.
Okay. One more general and then more specific comment. The 4D spatial model, tetrahedral synergetic geometry, you have space and time for dimensions spatially represented rather than the XYZ. You have three spatial dimensions. And then there's like the special question of how does T come into play? So some is known there and some is not known there, but when and where and how are, of course, very linked. And if we take model-based science seriously, then they're also very enabled and constrained by the kinds of models we have. So maybe it's related to how we think about geometry. And then the sense and action, are they continuous or are they discrete? We've seen recently DaCosta et al., like the synthesis of active inference on discrete state spaces. And then we've also seen some of Alex Shantz's work on the continuous control settings with like the mountain car, which is not just the discrete case. So it's just interesting to see how some of the theory goes out in front. And then there's parallel lines of mathematical development with continuous and discrete modeling, like digital and analog modeling. And sometimes those types of modeling scientifically are very well reconciled. And other times they result in pretty fundamentally different patterns. Like there are certain equations where the discretized form or the version that's computed without really high precision numbers on a computer has really different behavior than the other kinds of models, which gets at the importance of thinking about how the model is framed and then also applied. Because like if the precision of the processor matters for what the paper discovers, then shouldn't that be a part of the consideration?
Okay. Here's a few of the other comments for 36.1.2. So by the way, we'll write down any other questions. So if anyone in the last few minutes has some questions, they can ask. And we'll be just writing things down, getting excited for .2. And if anybody wants to join who wasn't here today, they're also very welcome to. How is representation related to autopoiesis? So refresher, autopoetic system is capable of producing and maintaining itself by creating its own parts. So how does representation differ or does it differ for a system that is just engineered and secreted into the niche, like a computer versus potentially something that is reassembling its parts continually? Why does that matter if it does?
Is that a thing to look at in point to Daniel? That sounds good. Yeah.

Yeah. I'm trying to push it on. But I mean, the reality is, is that we can take each of the different segments, the organizational, the structural, the content related, and then put them through the autopoetic grinder and see what pops out the other side. Okay. Great suggestion. Apply this to the four columns. What other things would be fun to discuss in .2 or what else can go through the four column grinder or what would be any other question that any of you have?

In the in the in the in the .2 also, we can, we can spend a little bit of time on some of the, some of the comments that were in the section five of the paper, specifically around how we face state from potentially, they didn't say that this is true. But they said, if this claim is true, we could be phase stating from non representational to representational. Maybe, maybe could we spend a little bit of time, not just looking at the autopoiesis aspect of that, but how does, how does that density form in the in relationship to this idea of modeling ourselves? Great question.

Here's a question that Stephen wrote. So feel free to give a thought on it, or we can push it.

It was, will robotics or neuro phenomenology help most in testing these complex temporal dynamics in a realist way?

I'm just going to say that's a great question. And I think we need the person who asked the question to come on and chat with us about that a little bit.

We're just delegating so hard here. Blue, go ahead.

So something that I've brought up, I think in the .0 is like, I would really like to tease apart the similarities and differences between like structural aspects and content related aspects. I'm not entirely certain, like, that's a fuzzy blurred line for me. There. So maybe we could get into that a little bit more.

I agree. It's fuzzy for me too.

These are awesome questions too. Thanks for writing this all out blue. Well, we have multiple pages.

So we have autopoiesis from non-rep to rep and back and forth or whatever else. How does that form in the context of self-representation? I think there's a few Hamlet quotes that'll come into play there.

Then Stephen's question about technology and realism.

Structural and content aspects. Detached representations in the neurobehavioral systems in the sensory motor loop.

Niche modification. Stigma G and the digital case, extended cognitive systems that include digital and mirror adaptive systems. Stephen singing 80 songs in the chat, I believe.

Thinking through other minds and multi-scale integration. One of Sims's other papers.

How are they related? Does the FEP exist independently or only in the mirror?

A lot of ways to answer that one.

One part we could look into now or maybe in the dot two would be the free expected free energy and variational free energy. And how they play different roles in the FEP.

These are also some things like if people are watching live or in the next few days, then they can write a comment on it or get in touch with us. Or like if they want to submit a question or they can come on and discuss it in the dot two. Variational and expected free energy.

There's a lot of parts to the paper.
And action oriented representation. We didn't get to that slide in the dot zero.
But just what is action oriented representation? We've talked about representations today, but then did we talk about, were we implicitly talking about action oriented representations?
Are those a subclass? How are they different?
Any other? Yeah, blue, blue. In our system, in human cognition, there are some hard coded representations that we have. Action oriented, maybe one of them, like kinesthetic memory and also visual recognition systems. And it's really interesting that, and perhaps easy to apply the FEP to specific cognitive systems, like subsystems, but it's really much more difficult when we have like multiple competing processes. Like we're not just playing the piano and we're not just looking at the piano keys. Like we're doing both of those things at the same time and lots of other things. Like we might be, you know, having some memory of one time that we heard the song that we're playing or something like that. So it's really interesting to think about representation and all the ways that it exists. Also, even in memory, right? Like the memory representation is, it's like a, it's not, that's not a hard coded thing. It's really squishy and it's subject to change through time. And, you know, we might remember a situation, like you can tell me your phone number right now and I'll remember it, you know, in five minutes, but in five days, no, like, or something like that. So I don't know. It's really interesting to think about the FEP and how it can apply in this overarching way where there are multiple competing goals and priorities and yeah, that.
One thought I hope I'm not misrepresenting an empirical result was in fruit fly, the Drosophila.
They found that short-term memories didn't require protein synthesis, like just second to second, some sound or some symbol could be associated with a preferred or repulsive outcome, but preventing protein synthesis blocked the consolidation of memory and perhaps that's similar to other systems. And so there's kind of like the resonant neural structures that are just using the system's connections as they are. That's sort of one, perhaps faster mode of cognition that's like running around figure two.
And then there's a restructuring that requires changing the model, like structure learning, but that also connects to realism because we're talking about the synapses and the neurogenesis, like of specific cells that could be maybe measured. And then the neurophysiology and the behavior can be observed as well. If we have integrative frameworks like act-inth. So like it also relates to Dean's comment about the scales, the timescales of change. Like neural circuits cannot be changing at a timescale faster than neural firing. So how does the spiking component or the other sensory transduction or other chemical components, how do those overlapping timescales coordinate?
Dr. This reminds me too, Daniel, of something that we talked about like the very first time we met, because it was a recent experiment at that time about the memory transfer between snails cells, via injection of RNA. So that was like an interesting possible memory storage. And I'm not sure what else has, you know, developed from that, but also we've talked about, you know, ants and what, how they leave, like, they like, you told me like that they're just throwing up RNA, like they're throwing up all the time all over the place and like maybe leaving chemical like RNA traces for each

other. We talked about that too. So maybe, I don't know if you want to expand on that or, or does that tie into this protein synthesis, you know, thing?
Dr. One, I thought on that it was a recent ant paper where basically the act of mating, which we won't describe on this family friendly discussion, directly infuses neurotransmitters into the recipient and that changes their behavior and induces all these other hormonal changes. And so the realism of cognition, like the embodiment and the physicality of these systems, it is perturbable by infusions of neurotransmitters in other neural systems and in us. So that's sort of where realism can't be eradicated amidst all of these representations because things do hit our head and molecules do diffuse. It's kind of like grounding it. And that's, I think why it's such an interesting and integrative area with a lot of contention because the 4E side is holding down that real side.
And then the science side, so-called, is holding down the abstractions and the representations.
And then cognitive science, especially as applied to humans, is kind of where it meets in the middle.
It's like, well, yeah, you're talking about embedded cognition and inactive cognition, but that's your cognition. That's modeling. And so then you get all kinds of fun of how mixed and complex that area is.
Thanks, Blue and Dean and those who participated in chat for this really great discussion. I hope we can digest it and learn in the next week and reflect on some of these questions and anyone can ask more questions or start to prepare some thoughts so that they can also be involved because this was really fun.
Yeah, thank you all.
streams. I don't know ultimately how it's decided which order that we do them in, but the fact that we've gone from Connor's paper to Majee's paper to this one, whether it was done with some sort of intent and looking back, allow back to the future, or it was just, okay, well, I think we'll do this one next and see what happens. I think the order of these last three papers in particular has been pretty, just the order in which we've looked at them has generated a whole bunch of interesting ideas, including sort of the idea of intransitivity in the last paper now potentially influencing how we've discussed this paper. So again, I don't know how much pre-thought goes into that, but if there was some pre-thought that went into putting these papers in this particular order, it's been very helpful. And one other thing, if I find something happy because it's true, I'm going to laugh.
And if I find something happy because it's playing a game like 20 questions or any game, I'm going to laugh. So if I can't be, if I can't enjoy and really appreciate the sophistication of the conversation here, I probably wouldn't show up. So thanks to both of you.
You can make me read active papers, but you can't make me have fun.
But just to give it one closing answer, information for anybody who wants to like get involved with the discussions. Here's our coda for the dot coms unit. And so here's the live stream table, the one that gets represented on the viewable public one. So we have all the papers planned, like in this case, up till 37, 38, 39, 40, up through the end of March. So we try to plan the papers in advance so that people can have time to read them and commit to collaborating on the dot zero background video.
And then we have papers to potentially discuss. And so here people add papers and then they click a heart

and Dean, we just sorted by hearts. And then we picked the papers that were on the four top papers.
And we said, okay, we're going to read these four for 36, seven, eight, nine.
Yeah, those. And then we emailed the authors. And so we have different papers and we said,
well, what dates might work? Because we always will prefer to schedule a paper that we want to read
when the authors want to join. And so that's kind of our first, like the way that we pull papers out,
like we set the dates for the quantum free energy discussions in March. And then we work backwards
from there and then just try to not do two, not even overlapping papers, just try to provide
rhythm and novelty and philosophy and technical details. So it's sort of a local decision-making
based upon the votes that people provide who are participating in the dot coms unit.
And pretty random.
Yeah. And just what people service.
I was going to say, that's a pretty serendipitous order then. That's fantastic.
It's fine.
And if anybody watching or listening wants to suggest a paper for the live stream,
please do so.
Yeah. All right. Thank you, Dean and Blue. So see you all next week for the dot two. Bye.
Bye.

Thank you.